



Project Documentation

Sphere Coding Club — Project 4



Project title: Weather App

Team Name: Electron Edge

Team Members:

- ❖ **Mihir Mendake — Backend & Documentation**
- ❖ **Vaibhav Bandgar — Frontend**
- ❖ **Om Shinde — Backend & Presentation**
- ❖ **Ayush Teli —**

Submission Date: 07 February 2026

Submitted To: Sphere Coding Club

Project Type: HTML & CSS, API Calling Web Application

1. Introduction

Weather forecasting applications play an important role in daily life by helping users plan activities based on current and upcoming weather conditions. With the advancement of web technologies, it is now possible to build fully functional weather applications that run entirely in the browser without requiring a backend server.

SkyCast is a responsive, web-based weather application that retrieves real-time weather data using a public weather API directly from the browser. The project focuses on delivering an immersive user interface, smooth animations, and reliable weather information while remaining lightweight and easy to deploy.

2. Objectives

The objectives of this project are:

- To build a modern and visually appealing weather application
- To fetch live weather data using an API without implementing a backend
- To display essential weather details in a clean dashboard format
- To ensure smooth performance on both mobile devices and laptops

- To improve user experience through animations and responsive design
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3. Tools and Technologies

Tool / Technology	Purpose
HTML5	To create and structure the web pages.
CSS3	To style the pages, manage layout, and add visual elements.
Javascript (JS)	API calling, Handling Logic, and updating UI
Netlify	For deployment

4. Project Overview

The Weather App allows users to search for any city and view its current weather conditions along with additional details. The application communicates directly with a weather API using JavaScript, making it a front-end-only solution.

The application displays:

- Current temperature and weather condition
- City name and date
- Humidity, wind speed, UV index, and visibility
- Sunrise and sunset timings
- A 7-day weather forecast

Because all API requests are made from the browser, the application can be hosted easily on any static hosting platform.

The layout is fully responsive, ensuring consistent appearance and usability on both mobile devices and laptops.

5. Daily Progress Summary

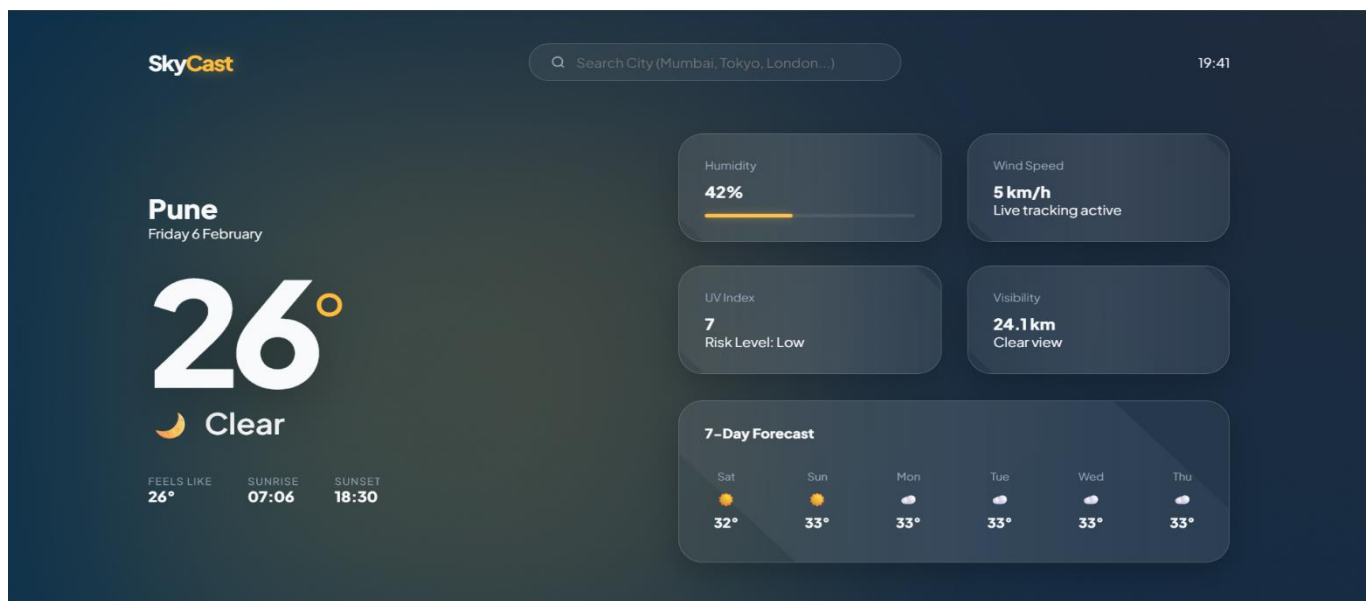
Date	Task Completed	Members involved
22 jan	First team meeting and task distribution	All members of electron edge
23 jan	Frontend started	vaibhav
24 jan	Search for public Api's	Vaibhav,Om,Mihir
25 jan	Frontend and its integration is done , Testing of Api	Vaibhav,Om,Mihir
26 jan	Debugging of the problems regarding Api	Om
27 jan	New public Api is used which calls from browser so no need of backend	Om, Mihir, Vaibhav
28 jan	Little tweaks and testing	Om, Mihir, Vaibhav
29 jan	Documentation work started	Mihir
30 jan	PPT work and final testing	Om
31 jan	Work in progress	Om , Mihir
01 feb	Ppt and documentation work done	Om, Mihir, Vaibhav
02 feb	Debugging and final testing	Om, Mihir, Vaibhav
07 feb	Submission day	Om,Mihir,Vaibhav

Navigation: Here is the deploy link of the app.

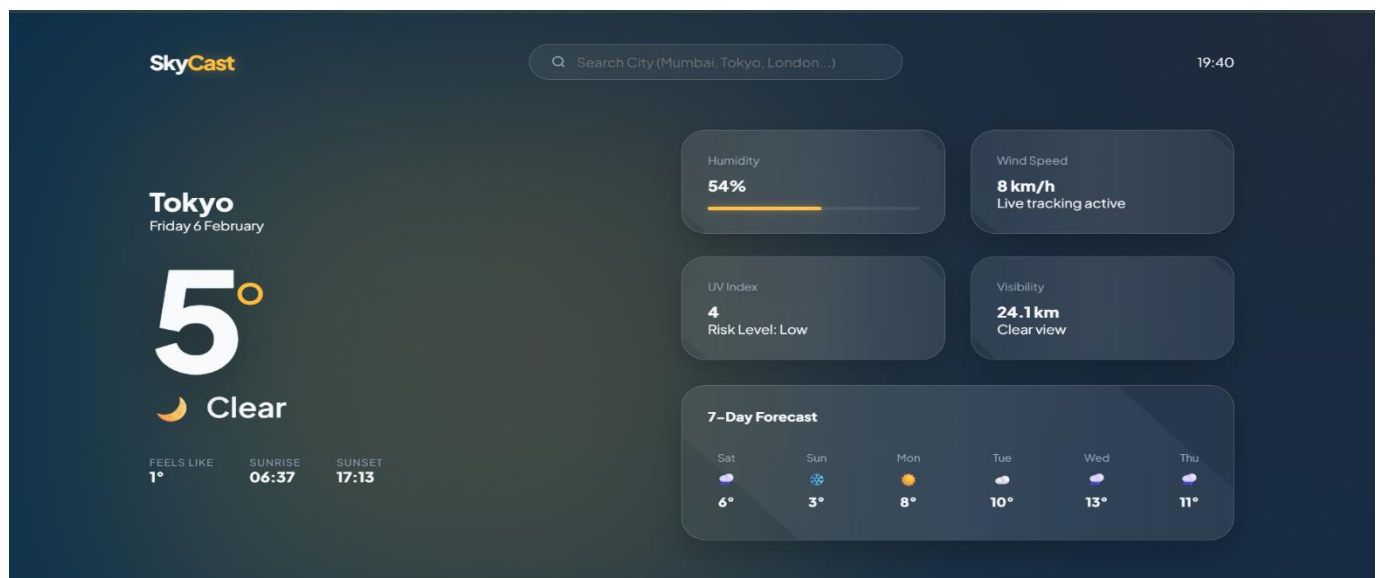
Deploy link: <http://skkycast.netlify.app/>

6. Screenshots

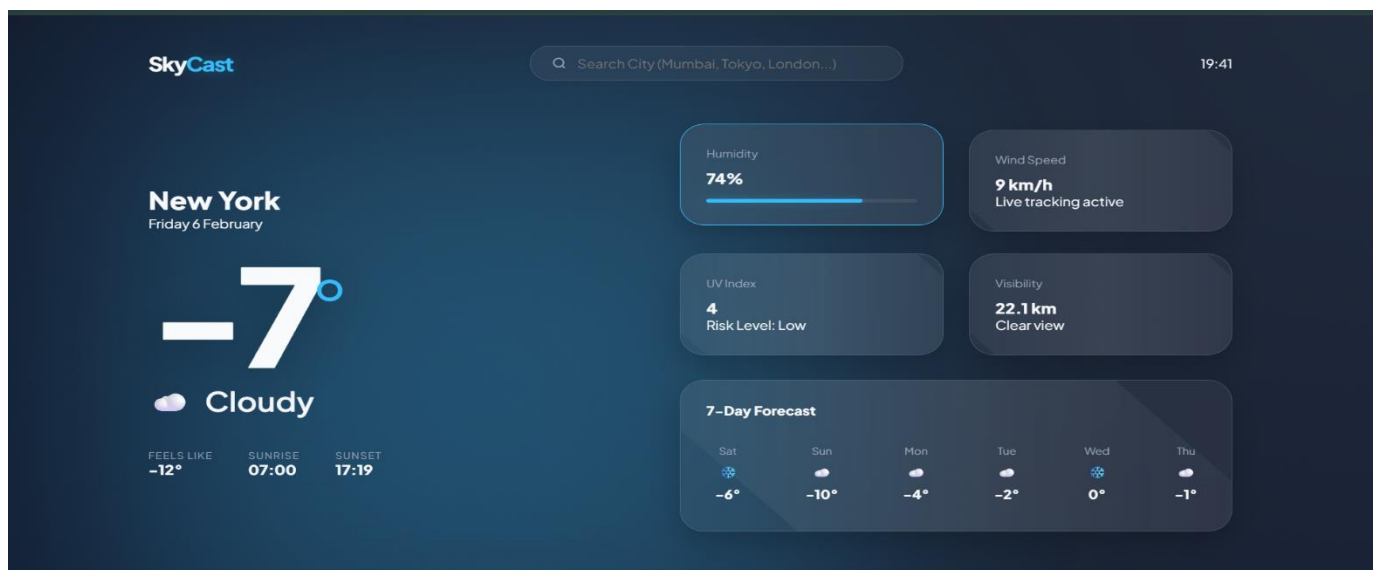
Pune:



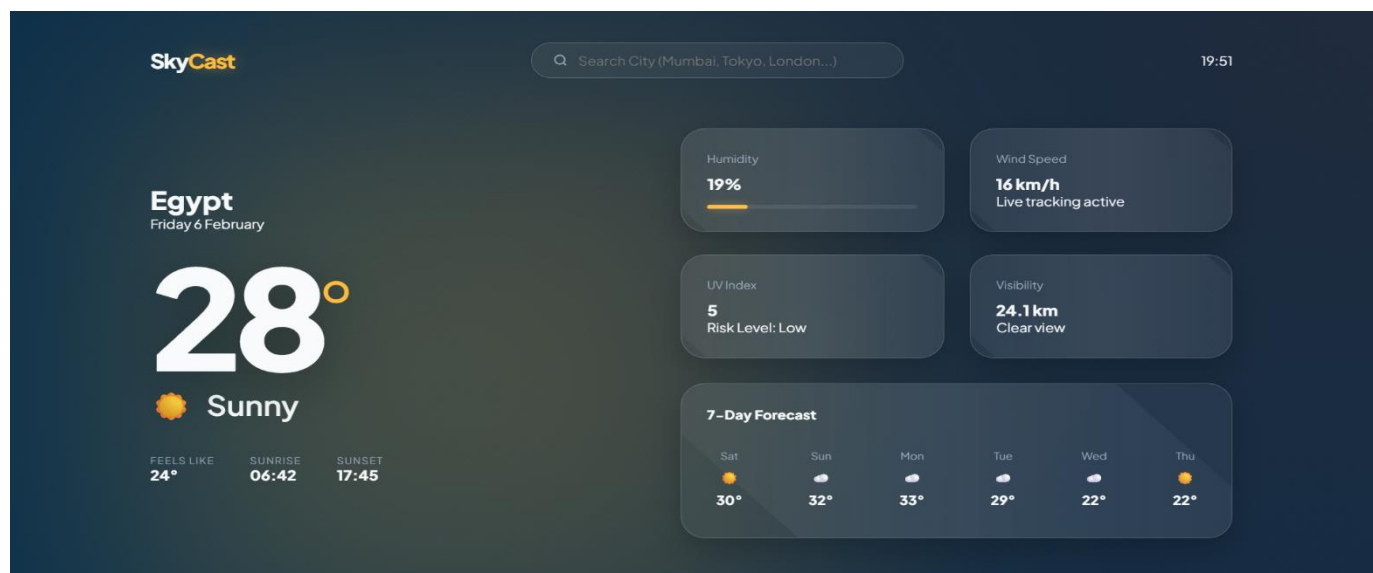
Tokyo:



New York:



Egypt:



7. Learning and Challenges

Learning Outcomes:

- Understanding how APIs can be consumed directly from the browser
- Working with asynchronous JavaScript operations

- Designing a fully responsive user interface
- Using the Canvas API for animations

Challenges Faced:

- Handling API responses and error cases
 - Maintaining performance while using animations
 - Ensuring consistent UI behavior across screen sizes
 - Managing API keys in a front-end-only project
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8. Conclusion

The Weather App project successfully demonstrates how a functional and responsive web application can be built using only front-end technologies. By fetching weather data directly from an API through the browser, the application eliminates the need for backend development while still providing accurate and dynamic weather information.

9. References

- Claude Ai
 - Youtube – weather Api
 - Xiaomi weather app
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- Final note

This project highlights the effectiveness of modern front-end development. The Weather App serves as a strong foundation for future improvements and demonstrates the team's understanding of UI design, API integration, and responsive web development.

— **Team Electron Edge (Mihir | Om | Vaibhav | Ayush)**

— **Sphere Coding Club**